

Youssef Shalaby

ML Engineer

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Profile

Data Scientist and AI Engineer with **1+ year of hands-on experience** building machine learning and deep learning solutions. Skilled in Python, Scikit-learn, neural networks, and data analytics, with practical experience in real-world ML projects and AI-driven problem solving. Passionate about transforming data into actionable insights and continuously advancing in AI technologies.

Education

Bachelor of Computer Science, Arab Open University 09/2023 – present

- GPA: 3.65/4.0

Professional Experience

Machine Learning Engineer Intern, Elevvo Pathways 10/2025 – 11/2025

- Applied machine learning techniques to real-world datasets, including data preprocessing, feature engineering, and model evaluation. Remote
- Built and evaluated supervised and unsupervised learning models for tasks such as student score prediction, customer segmentation, forest cover classification, and loan approval prediction.
- Used Python, Pandas, Scikit-learn, Matplotlib, and XGBoost to develop end-to-end ML pipelines.

AI Intern (Generative AI – NVIDIA DLI Summer Training), 08/2025 – 09/2025

Information Technology Institute (ITI) Hybrid

- Completed an intensive **35-hour NVIDIA Deep Learning Institute (DLI) Summer Training Program in Generative AI.**
- Designed and implemented **LLM-based applications** using **prompt engineering** techniques.
- Enhanced large language models using **Retrieval-Augmented Generation (RAG)** for improved accuracy and contextual responses.
- Earned **NVIDIA Certificates of Competency & Completion** in:
 - Building LLM Applications with Prompt Engineering
 - AI for All: From Basics to GenAI Practice

Data Scientist Trainee, Digital Egypt Pioneers Initiative (DEPI)– Ministry of 11/2024 – 03/2025

Communications and Information Technology Remote

- Completed intensive training in data science and machine learning as part of a national program sponsored by MCIT.
- Developed end-to-end data science workflows, including data cleaning, EDA, feature engineering, modeling, and interpretation.
- Applied supervised and unsupervised learning techniques to real-world case studies and datasets.

Projects

AI Health Assistant – Multi-Disease Prediction System [↗](#)

Developed a **Streamlit-based ML web app** that predicts the risk of **Diabetes, Heart Disease, and Parkinson's Disease**. Trained and compared multiple models and deployed the best ones for real-time predictions.

Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy, Streamlit

Loan Approval Prediction [↗](#)

A machine learning classification project that predicts whether loan applications are approved or rejected. It includes: data cleaning, EDA, training & comparing models like Random Forest and XGBoost, and evaluation with metrics.

Covertypes Forest Cover Prediction [↗](#)

Multi-class classification project using the UCI **Covertypes dataset** to predict forest cover types via models such as Random Forest and Gradient Boosting. Demonstrates data preprocessing, model training, and evaluation on a real benchmark dataset.

Customer Segmentation [↗](#)

Applies **K-Means clustering** to segment customers based on demographic features. Good example of unsupervised learning and exploratory data analysis

Volunteer

Microsoft Student Club, *Instructor of Problem solving club*

02/2023 – 06/2023

- Led problem-solving sessions for students, enhancing their analytical and coding skills.
- Developed and delivered engaging lessons, improving public speaking and instructional abilities.
- Gained leadership experience by organizing and managing training sessions.

Skills

Machine Learning

- Supervised & Unsupervised Learning (Regression, Classification, Clustering)
- Model Evaluation, Feature Engineering, Hyperparameter Tuning

Web Scarping

Beautiful soup, Selenium

Programing Language

Python, c++

Data Processing & Analysis

Numpy, Pandas

Data Visualization

Matplotlib, Seaborn, Plotly

Model Deployment

Streamlit, FastAPI, Hugging Face Spaces

Certifications

- AI for All: From Basics to GenAI Practice, Nvidia [↗](#)
- Practical Machine Learning for Data Science, ITI [↗](#)
- AI & Machine Learning Foundation, Sprints [↗](#)
- Python Essentials 1, Cisco [↗](#)
- Generative AI from Google [↗](#)
- Data Visualization with Python, IBM [↗](#)
- Python Programming Basics, ITI [↗](#)
- Building LLM Applications With Prompt Engineering, NVIDIA [↗](#)